

Momentum

Linear Momentum (p) is a measure of the quantity of motion of a moving body.

- **Equation:** $p = m \times v$
- **SI Unit:** kilogram meter per second (kg m s^{-1} or N s).
- **Key Fact:** Momentum is a **vector quantity**. It has both magnitude and direction.

Conservation of momentum: total momentum before = total momentum after, provided no external resultant force acts on the system (closed isolated system).

- **"System":** The group of objects you are considering.
- **"External resultant force":** Forces from outside the system (e.g., friction, gravity if objects are large). Internal forces (objects pushing on each other) do not change the total momentum.
- This principle applies to **all collisions, explosions, and Jet propulsion / rocket motion**.

Force and Momentum: The Fundamental Connection

Newton's Second Law says:

“The net force acting on a body is equal to the rate of change of its linear momentum.”

Therefore:

$$F = \frac{\Delta p}{\Delta t}$$

Newton's Third Law says:

“When two bodies interact, the force on body A from body B is equal in magnitude but opposite in direction to the force on body B from body A”

$$F_{A \text{ on } B} = -F_{B \text{ on } A}$$

How This Leads to Momentum Conservation

For two interacting bodies A and B:

$$F_{A \text{ on } B} = -F_{B \text{ on } A}$$

Apply Newton's Second Law to each:

- Force on B = rate of change of B's momentum: $F_{A \text{ on } B} = \frac{\Delta p_B}{\Delta t}$
- Force on A = rate of change of A's momentum: $F_{B \text{ on } A} = \frac{\Delta p_A}{\Delta t}$

Put these together:

$$\frac{\Delta p_B}{\Delta t} = -\frac{\Delta p_A}{\Delta t}$$

Multiply by time (same Δt for both):

$$\Delta p_B = -\Delta p_A$$

$$\Delta p_A + \Delta p_B = 0$$

This means: The total change in momentum is zero → **Total momentum is conserved.**

What This Means for Collisions

During any collision:

- the forces on the two bodies (or on A and B) are equal and opposite (Newton 3rd)
- force equal/is proportional to rate of change of momentum (Newton 2nd)
- The time is the same for both forces hence change in momentum is the same

Momentum is always conserved (in a closed system of course) so when an object collides with the wall or with the ground even though the momentum of object itself reduce and the ground/wall might not have noticeable motion (there actually is but its too small to notice) but the momentum is transferred to the ground/wall or even earth.

the reason the motion is small is because relatively the mass of earth is too big meaning the velocity is too small (negligible)

One-Dimensional Momentum Problems

For problems in one line (e.g., along a straight track), we use a **sign convention**.

General Conservation Equation:

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

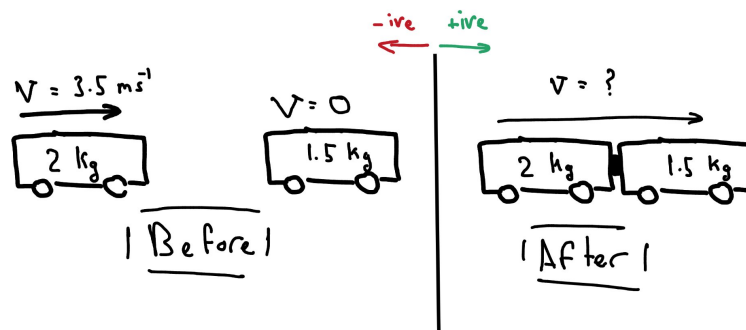
(where u = initial velocity, v = final velocity)

Rule: ALWAYS define your positive direction before starting a calculation.

Type 1: Collisions

Worked Example 1 (Inelastic Collision - Objects Stick Together)

A trolley of mass 2.0kg moving at 3.5ms^{-1} collides with a stationary trolley of mass 1.5kg . They stick together and move off. Calculate their common velocity.



Step-by-Step Solution:

Equation Setup:

Total momentum before = Total momentum after

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v$$

Substitution & Solve:

$$(2.0 \times 3.5) + (1.5 \times 0) = (3.5) \times v$$

$$7.0 + 0 = 3.5v$$

$$v = \frac{7.0}{3.5} = 2.0\text{ms}^{-1}$$

Answer with Direction:

$v = 2.0\text{ms}^{-1}$ in the original direction of motion (positive).

Type 2: Explosions

Objects are initially together and stationary. They separate due to an internal force (e.g., a spring). Total momentum before an explosion is **zero**.

Equation from Conservation

$$0 = m_1 \vec{v}_1 + m_2 \vec{v}_2$$

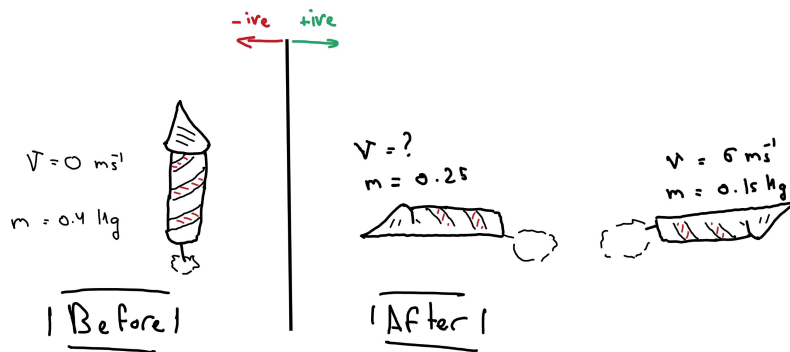
Therefore:

$$m_1 \vec{v}_1 = -m_2 \vec{v}_2$$

(The momenta are equal in magnitude but opposite in direction).

Worked Example 2 (Explosion)

A stationary firework of mass 0.40 kg explodes into two pieces. Piece A (0.15 kg) flies off to the right at 6.0 ms^{-1} . Calculate the velocity of piece B (0.25 kg).



Equation:

$$0 = m_A v_A + m_B v_B$$

Substitute:

$$0 = (0.15 \times 6.0) + (0.25 \times v_B)$$

$$0 = 0.9 + 0.25v_B$$

$$v_B = \frac{-0.9}{0.25} = -3.6\text{ ms}^{-1}$$

Answer: Velocity of piece B is 3.6 ms^{-1} **3.6 m s⁻¹ to the left.**

Elastic vs. Inelastic Collisions

Collisions are classified by what happens to **kinetic energy** (E_k).

- **Elastic Collision:** Total kinetic energy is conserved. (KE before = KE after)
- **Inelastic Collision:** Total kinetic energy is **NOT** conserved.

Definition Table

Feature	Elastic Collision	Inelastic Collision	Perfectly Inelastic
Kinetic Energy	Conserved $E_{k,before} = E_{k,after}$	Not conserved $E_{k,before} > E_{k,after}$	Maximum loss Objects stick together
Momentum	Conserved	Conserved	Conserved
Energy Transformation	None (ideal)	→ Heat, sound, deformation	→ Heat, sound, deformation
Relative Speed Rule	Speed of separation = Speed of approach	Speed of separation < Speed of approach	Minimum speed of separation
Examples	Billiard balls, gas molecules (approx.)	Car crashes, putty hitting wall	Two objects sticking together

The Kinetic Energy Rule

Definition:

- **Kinetic Energy before** = $E_{k,before} = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2$
- **Kinetic Energy after** = $E_{k,after} = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$

Rule:

$$E_{k,before} = E_{k,after}$$

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

The Relative Speed Rule

Definition:

- **Speed of approach** = $|u_1 - u_2|$ (before collision)
- **Speed of separation** = $|v_2 - v_1|$ (after collision)

Rule:

$$|v_2 - v_1| = |u_1 - u_2|$$

where u_1, u_2 are initial velocities and v_1, v_2 are final velocities.

Absolute value (“|””) was used to make sure the values for relative speed of approach/ separation are always positive

Two-Dimensional Momentum

In 2D, momentum is conserved **independently in perpendicular directions** (e.g., x and y).

Method:

1. Resolve all velocities into components.
2. Apply conservation of momentum **separately** for the xx -components and the yy -components.
 - $\sum initial p_x = \sum final p_x$
 - $\sum initial p_y = \sum final p_y$
3. Combine the final xx and yy momentum components to find resultant velocities.

Worked Example 3 (Oblique Collision with Stationary Object)

A ball A (mass m) moving at 5.0 m s^{-1} East collides with a stationary identical ball B. After collision, ball A moves at 3.0 m s^{-1} at 30° North of East. Find the velocity of ball B.
(Take East as positive x , North as positive y .)

initial Momentum:

$$p_x(\text{initial}): m \times 5.0 + m \times 0 = 5.0m$$

$$p_y(\text{initial}): 0 + 0 = 0$$

Resolve Final Velocities:

$$A: v_{Ax} = 3.0 \cos 30^\circ \approx 2.60 \text{ m s}^{-1}, v_{Ay} = 3.0 \sin 30^\circ = 1.50 \text{ m s}^{-1}$$

$$B: v_{Bx} = ?, v_{By} = ? (\text{Let velocity be } v_B \text{ at angle } \theta)$$

Apply Conservation:

$$x - \text{direction}: 5.0m = m(2.60) + m(v_{Bx}) \Rightarrow v_{Bx} = 5.0 - 2.60 = 2.40 \text{ m s}^{-1}$$

$$y - \text{direction}: 0 = m(1.50) + m(v_{By}) \Rightarrow v_{By} = -1.50 \text{ m s}^{-1}$$

Find Magnitude & Direction of $\vec{v}_B$:

$$\text{Magnitude: } v_B = \sqrt{(2.40)^2 + (-1.50)^2} = \sqrt{5.76 + 2.25} = \sqrt{8.01} \approx 2.83 \text{ m s}^{-1}$$

$$\text{Direction: } \theta = \tan^{-1} \left(\frac{|-1.50|}{2.40} \right) \approx 32^\circ \text{ South of East (since } v_{By} \text{ is negative)}$$

Force on Fluids: Mass Flow Rate

Mass Flow Rate

Mass flow rate ($\dot{m}$) is the mass of fluid passing a point per unit time.

$$\dot{m} = \frac{\Delta m}{\Delta t} \text{ or } \dot{m} = \rho A v$$

Where:

- $\dot{m}$ = mass flow rate (kg s^{-1})
- ρ = density of fluid (kg m^{-3})
- A = cross-sectional area (m^2)
- v = flow velocity (m s^{-1})

Force from Fluid Flow

$$F = \rho A v^2$$

Where:

- F = Force on fluid (N)
- ρ = Density of fluid (kg/m^3)
- A = Cross-sectional area (m^2)
- v = Flow speed (m/s)

Note: this formula can only be used When fluid accelerates from rest OR decelerates to rest

Worked Example 4: Water Jet on Wall

Water of density $\rho = 1000 \text{ kg m}^{-3}$ emerges from a hose of area $A = 4.0 \times 10^{-4} \text{ m}^2$ at $v = 15 \text{ m s}^{-1}$ and hits a wall perpendicularly. Calculate the force on the wall.

Apply the formula

$$F = \rho A v^2$$

Substitute values

$$F = 1000 \times (4.0 \times 10^{-4}) \times (15)^2$$

Calculate

$$F = 1000 \times 4.0 \times 10^{-4} \times 225$$

$$F = 90 \text{ N}$$

Interpretation

- Force **on water** by wall = -90 N (opposite to flow)
- Force **on wall** by water = $+90 \text{ N}$ (in direction of initial flow)